

Case Study – Letterboxd Sentiment Analysis

General Description:

Submit a link to a GitHub repository including all inputs and outputs necessary for completion of the project and creation of deliverables.

Why am I doing this?

Sentiment analysis is one of the most widely applied techniques in data science, powering everything from product review monitoring to social media trend detection. The core idea is straightforward: given a piece of text, classify it as positive, negative, or neutral, but the choice of model matters a lot. A rule-based approach like VADER, which scores words using a lexicon, can behave differently from a transformer like HuggingFace, which learns on millions of social media posts. This case study gives you the opportunity to run both models on the same dataset of real Letterboxd movie reviews, evaluate their outputs against a ground truth, and draw your own conclusions about which approach works better and why.

- Course Learning Objective: Apply and evaluate pretrained NLP sentiment models on real-world text data

What am I going to do?

You will work with a dataset of 4,755 Letterboxd reviews from the platform's Top 250 films. Each review has a written text and a star rating. The star rating is your ground truth, mapped to Negative (≤ 1.0 stars), Neutral (1.5–3.0), or Positive (≥ 3.5). HuggingFace RoBERTa predictions are pre-computed in the dataset so you do not need a GPU. Your job is to run VADER on the same reviews, evaluate both models using accuracy and Macro-F1, produce comparison figures, and organize everything into a clean repository. A starter script is provided that handles the data loading and VADER scoring, your job is to extend it through the evaluation and figures.

Tips for success:

- Read articles before starting: one explains VADER, the other covers HuggingFace
- Use the starter script. Steps 1–3 are fully implemented. Start there and build outward.
- Pay close attention to the class imbalance. Accuracy alone will be misleading, so use Macro-F1 as your primary metric and explain why in your repo.
- Look at the reviews the models get wrong. Failures are more informative than successes.
- Ask others (Professors, TAs, classmates, friends, etc) for help. 2 minds are better than 1.

How will I know I have Succeeded? You will meet expectations on Case Study – Letterboxd Sentiment Analysis when you follow the criteria in the rubric below:

Spec Categories	Spec Details
Formatting	• Repository: A GitHub repo containing all materials. ◦ Scripts

	○ Data ○ Figures ○ Articles ○ README.md
Scripts	● Goal: This folder contains all source code for your project. ● Include all scripts used. Name each script to make clear what it does and what order to run things. ● Each script should begin with a short header comment explaining its purpose and how to run it and inline comments throughout explaining what each section of code accomplishes and why. ● At minimum your Scripts/ folder should contain your analysis script, which runs VADER, evaluates both models, and generates all figures.
Data	● Goal: This folder contains all data used in the project. ● letterboxd_reviews.csv is provided. ● Your analysis script will produce <i>letterboxd_reviews_with_vader.csv</i> to include in the Data/ as well. ● Include a brief Data Appendix as a PDF describing: ○ What the unit of observation is ○ A short description of each key variable used in the analysis. ○ The distribution of true_label and a note on the class imbalance.
Figures	● Goal: This folder contains all output figures generated by your analysis. ● Use informative, descriptive file names. ● Required figures: ○ A grouped bar chart comparing accuracy and Macro-F1 for VADER vs. HuggingFace. ○ Confusion matrices for both models, clearly labeled. ○ At least one additional figure of your choice ● All figures must have a title, labeled axes, and a source note.
README file	● Goal: Helps anyone who visits the repository understand what it contains and how to reproduce the results. ○ Section 1: Software and Platform ○ Section 2: Repository Map ○ Section 3: How to Reproduce Results ● Include a brief summary of key results (VADER vs. HuggingFace accuracy and Macro-F1) so a reader knows what to expect before running anything.
References	● Include all references used that were not already given in an individual reference file.